A Late Cretaceous paleomagnetic pole from Koto rhyolite, southwest Japan: Implications for eastern margin deformation of Asia

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[1] A paleomagnetic pole was obtained from Late Cretaceous rhyolitic ash flow tuffs and associated ring dikes, which are collectively called the Koto rhyolite in the eastern part of southwest Japan. Characteristic remanent magnetizations were isolated from 33 sites through thermal demagnetization. Normal and reversed polarity magnetizations were found and are mutually antipodal. Although the magnetization directions become more dispersed after tilt correction, the clearly defined polarity sequence and the positive contact test between ash flow tuffs and ring dikes ensure that the characteristic magnetizations of ash flow tuffs predate the intrusion of associated ring dikes. Since no regional tilting is expected, we regard the in situ magnetization directions to represent the Late Cretaceous (70 Ma) paleomagnetic field ($D = 84.6°$, $I = 57.5°$, $\alpha_{95} = 5.5°$) in eastern southwest Japan. The paleomagnetic pole is not significantly different from the coeval poles of western southwest Japan. This result gives a further support to a microplate model for the Miocene clockwise rotation of southwest Japan. By canceling out the effect of the Miocene rotation, we were able to obtain a paleomagnetic pole of 70 Ma (219.7°E, 63.0°N, $\alpha_{95} = 12.2°$) for southwest Japan when the Japan Arc was incorporated in the eastern margin of Asia. Paleomagnetic poles of Late Cretaceous age from Korea, south China coast, and restored southwest Japan are statistically indistinguishable from each other but significantly differ from that of the continental interior of east Asia. The observed rotation of about 20° probably results from shear deformation exerted by northeast-trending strike-slip faults along the eastern continental margin of Asia. *INDEX TERMS:* 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes; 1532 Geomagnetism and Paleomagnetism: Reference fields (regional, global)

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1. Introduction

[2] The eastern continental margin of Asia is an important field for testing the relative importance of plate subduction and extrusion due to continental collision, which occurred in the Late Mesozoic and Cenozoic. Structural features of this region are characterized by many strike-slip faults trending parallel to the continental margin as represented by the Tanlu fault in north China (Figure 1). Oblique subduction of the Kula or Pacific oceanic plates could cause tectonic movements along these strike-slip faults [e.g., *Maruyama et al.*, 1997; *Xu et al.*, 1987]. On the other hand, strike-slip faults can be regarded as a consequence of propagated deformation caused by India-Eurasia collision [e.g., *Peltzer and Tapponnier*, 1988].

[3] There are three possible models to explain continental margin deformation by tectonic rotation, as illustrated for the Pacific coast of North America [*Wells and Heller*, 1988]: (1) rotation of allochthonous terranes when accreted to a continent by oblique collision, (2) shear rotation of structural panels bounded by two major strike-slip faults, and (3) microplate rotation due to asymmetric intracontinental or back arc extension. On the opposite side of the Pacific, i.e., in the eastern margin of Asia; however, the tectonic deformation is still not well described based on reliable paleomagnetic data. There are two reasons for this situation. First, sufficient paleomagnetic reference data later than the Late Mesozoic had not been available for the continental interior of Asia. This is because previous paleomagnetic

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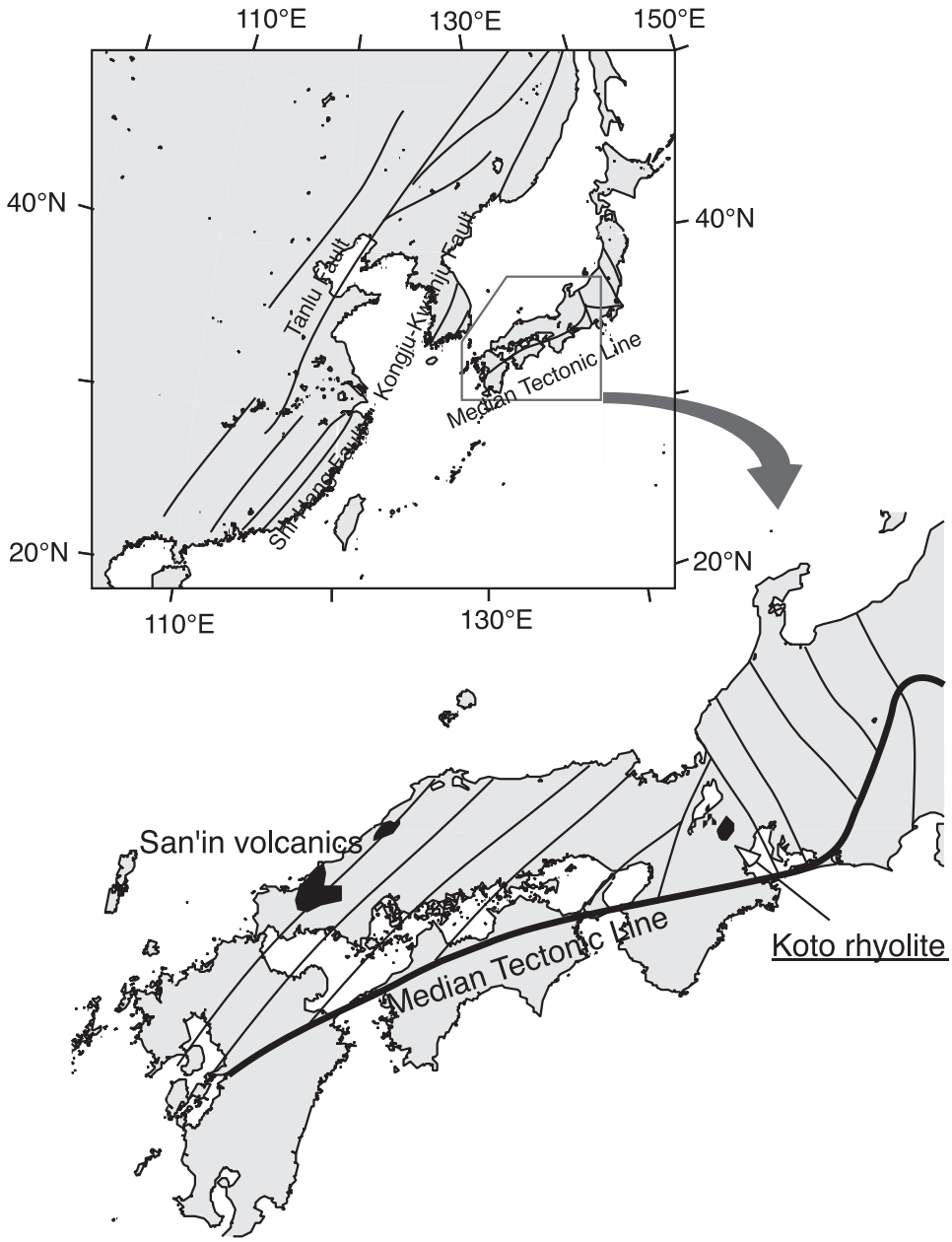


Figure 1. Simplified tectonic map of the eastern margin of Asia and southwest Japan modified after the work of *Kanaori* [1990], *Kimura et al.* [1990], and *Otsuki* [1992]. Locations of the Koto rhyolite and San'in volcanics are shown.

studies had mainly focused on the assembly history of east Asia prior to the late Mesozoic [e.g., *Zhao and Coe*, 1987]. Second, a series of back arc basins of Tertiary age between Asia and the Pacific Ocean make the tectonic process more complicated [e.g., *Jolivet et al.*, 1994]. To address the continental margin deformation, it is necessary to separate tectonic effects due to back arc spreading from other effects.

[4] The Miocene rotation of the Japan Arc, resulting from back arc spreading of the Japan Sea, has been well documented by a large amount of paleomagnetic data, especially for southwest Japan [*Otofuji*, 1996]. A large (~50°) clockwise rotation in the middle Miocene is required based on the excellent paleomagnetic database collected from diverse rock types covering almost the whole of southwest Japan. Nevertheless, two review papers reached mutually incompatible conclusions on the mechanisms of the rotation of southwest Japan [*Jolivet et al.*, 1995; *Otofuji*, 1996]. *Otofuji* [1996] interpreted the paleomagnetic results to support a microplate rotation model in which southwest Japan is assumed to behave as a single coherent block during the Miocene. This model is consistent with the nearly parallel arrangement of pre-Neogene geologic terranes along an ancient subduction zone in southwest Japan [*Saito*, 1985]. On the other hand, a shear rotation model that is based primarily on structural analyses of shear zones in southwest Japan [*Altis*, 1999; *Kanaori*, 1990] is favored by *Jolivet et al.* [1995]. They attempted to explain the paleomagnetic data by slip rotations of parallel elongate panels, which

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accommodate the shear of two major strike-slip faults. They estimated that about 40% of the Miocene rotation can be attributed to the internal deformation associated to the shear, while admitting that the microplate rotation of southwest Japan would account for the rest of the rotation.

[5] In principle, it is not possible to distinguish between the above two models based on paleomagnetic data alone, if structural panels are placed in parallel throughout the rotated block because either large-scale tectonic rotation of a single microplate or small-scale internal reorganization of structural panels can result in a same amount of rotation observed by anomalies in paleomagnetic declinations. This is the case for an extensive region of southwest Japan, where paleomagnetic directions are consistent at first approximation in space and time both before and after the Miocene rotation [*Otofuji et al.*, 1985], and structural panels are systematically juxtaposed in parallel with northeast-southwest trends [*Kanaori*, 1990] (Figure 1). Localized tectonic rotations due to strike-slip faults were reported from surrounding regions of southwest Japan [e.g., *Ishikawa*, 1997; *Kodama et al.*, 1993; *Yamaji et al.*, 1999]. In order to determine the appropriate tectonic models, it is important to obtain paleomagnetic data from regions with different tectonic structures.

[6] The Koto rhyolite, which is a felsic igneous complex of the Late Cretaceous, is located in the eastern part of southwest Japan. The direction of shear zones is transitional in this area from a NE-SW trend in the west to a NNW-SSE trend in the east (Figure 1). Paleomagnetic data from the Koto rhyolite are indispensable for distinguishing between microplate rotation and shear rotation models.

[7] By restoring the effect of the Miocene rotation furthermore, we can obtain a paleomagnetic pole when the Japan Arc was at the eastern margin of the Asia continent prior to the opening of the Japan Sea. Comparison of paleomagnetic poles from the restored southwest Japan, Korea, and south China coast with that of the continental interior of east Asia can provide an insight into the deformation of eastern continental margin of Asia.

2. Geological Setting

[8] Felsic magmatism took place in a wide area along the eastern margin of the Asian continent in the late Mesozoic [e.g., *Takahashi*, 1983]. In southwest Japan, felsic volcanic and plutonic rocks covered or intruded pre-Cretaceous allochthons and formed a geological framework of the present Japan Arc. The Koto rhyolite, composed of rhyolitic ash flow tuffs and associated intrusive rocks, is one of these Mesozoic felsic igneous rocks, and it represents an effusive phase of a felsic volcano-plutonic complex surrounding Lake Biwa in the eastern part of southwest Japan (Figures 1 and 2). Granite batholiths are exposed surrounding the Koto rhyolite and constitute a felsic ring complex with a diameter of about 40–60 km which is interpreted to be a large Mesozoic cauldron [*Sawada and Itaya*, 1993].

[9] The Koto rhyolite is subdivided into two separately exposed areas: the main body is exposed on the western flank of the Suzuka mountain range, while smaller bodies are sporadically exposed as hills in an alluvial plain or islets in Lake Biwa [*Nishikawa et al.*, 1983]. The stratigraphy of the main body was established through a series of field surveys [*Harayama et al.*, 1989; *Ishida et al.*, 1984; *Miyamura et al.*, 1976]. Ash flow tuffs, which are moderately or densely welded, consist of two lithologic units (Yatsuoyama pyroclastics and Kaiwara ash flow tuff) separated by a lacustrine sediment layer (Table 1). Felsic intrusive rocks occur as dikes intruding along double ring faults (Inugami granite porphyry) or an intrusive phase closely associated with the ash flow tuff (Hatasho quartz porphyry). Ring dikes give rise to contact aureoles in the surrounding ash flow tuffs showing that the ring dikes are younger than the ash flow tuffs. The smaller bodies in the alluvial plain or Lake Biwa are composed of ash flow tuffs and felsic intrusive rocks, which are lithologically quite similar to those of the main body. At some outcrops of the smaller bodies, ash flow tuff units are found to be separated by a lacustrine sediment layer, but the stratigraphic correlation to the main body or among smaller bodies is poorly understood due to their isolated distribution.

[10] Ash flow tuffs of the Koto rhyolite suffered pervasive hydrothermal alteration [*Mimura et al.*, 1976]. Unaltered phenocrysts of mafic minerals and groundmass are rarely found, and they are mostly replaced by secondary minerals such as chlorite, calcite, and clay minerals [*Harayama et al.*, 1989]. Mosaic-feature groundmass and recrystallized biotites are observed in contact aureoles produced by the ring dikes, which is distinguishable from the nonbaked ash flow tuffs in the petrographical textures.

[11] Geochronological studies provide detailed thermochronological constraints on the Koto rhyolite and surrounding granites. Biotite K-Ar ages of the ash flow tuffs and the ring dikes of the Koto rhyolite range from 68 to 75 Ma, which are consistent with biotite K-Ar ages of the surrounding granites, and some minor dikes intruding the surrounding granites also show a similar range of biotite K-Ar ages [*Sawada and Itaya*, 1993]. Zircon fission track ages range from 60 to 70 Ma for the Koto rhyolite [*Ito*, 1989]. These geochronological data indicate that the Koto rhyolite was emplaced at about 70 Ma accompanied by the intrusions of surrounding granites and that in some areas minor igneous activity continued until about 60 Ma at elevated temperatures above 250°C (closure temperature for zircon fission track age) in conjunction with some dike intrusions. In other words, these data suggest that no igneous activity continued after 60 Ma, which gives a minimum age of the Koto rhyolite. Such a late-stage activity is probably associated with epithermal or hydrothermal activity in the cauldron as indicated by the secondary minerals.

3. Sampling and Laboratory Procedure

[12] Paleomagnetic samples were collected at 60 sites from the main body and the smaller bodies of the Koto rhyolite: 41 sites from ash flow tuffs, and 19 sites from intrusive rocks (Table 1 and Figure 2). Sampling sites were designed to cover all the stratigraphic units of the main body and a variety of isolated hills in the alluvial plain and islets in Lake Biwa. We collected hand samples, which were independently oriented and amounted to about seven on average at each site. One or more standard paleomagnetic specimens were cut from each hand sample in the laboratory.

[13] To determine structural attitudes, we measured orientations of eutaxitic structures of ash flow tuffs, which

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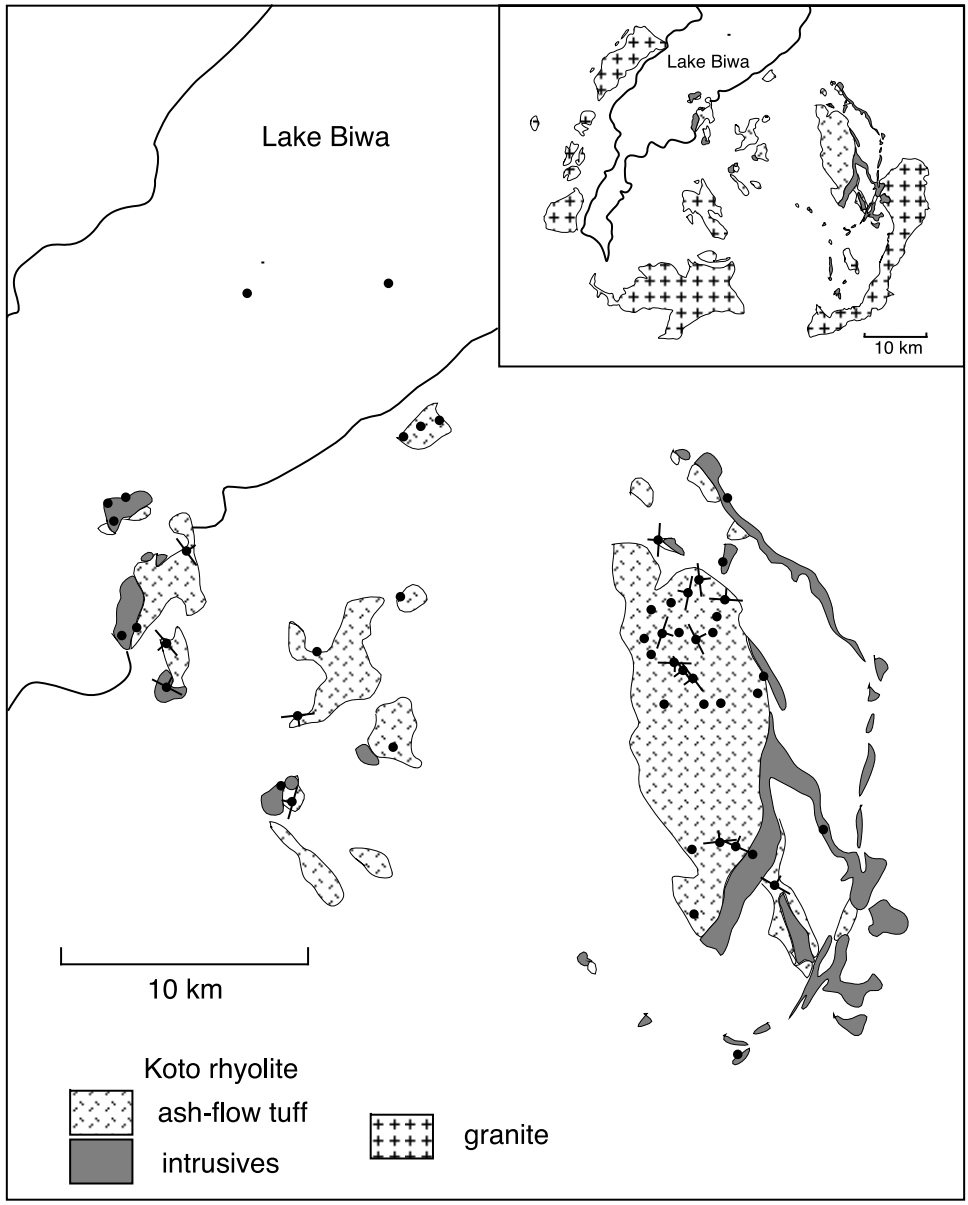


Figure 2. Simplified geological map of the Koto rhyolite and the surrounding coeval granite [*Sawada and Itaya*, 1993]. Sampling sites are shown with solid circles, and measured bedding attitudes are plotted for some sites of ash flow tuffs.

are represented by foliated pumices or lithic fragments. The measured attitudes are generally gentle; dipping angles of ash flow tuffs do not exceed 30° and do not show any regional trends (Figure 2 and Table 2), and our attitude data are concordant with those measured by *Harayama et al.* [1989], *Ishida et al.* [1984], and *Miyamura et al.* [1976]. It is known that eutaxitic structures do not necessarily represent ancient horizontal planes; for example, eutaxitic structures are parallel to the basement topography near the base of ash flows [*Hildreth and Mahood*, 1985]. Since such a nonhorizontal bedding is restricted to the base of ash flow tuffs, we avoided collecting samples near the base of ash

Table 1. Stratigraphic Relationships of Lithologic Units and Sampling Sites of the Koto Rhyolite for the Main Body and the Smaller Bodies Distributed in the Alluvial Plain or Lake Biwa

Main Body		Small Body	
Rock Unit	Site (KM)	Rock Unit	Site (KH)
Inugami granite porphyry	14, 24, 30, 31, 05, 15, 23, 32, 33	felsic intrusive rocks	05, 06, 07, 09, 10
Yatsuoyama pyroclastics	02, 16, 17, 19, 20, 27, 28, 34, 35, 36, 37, 38	Kamewariyama ash flow tuff	01, 02, 03, 04, 08, 11, 12, 13, 14, 15, 17, 19
Sediment layer		Koshigoe ash flow tuff	18
Hatasho quartz porphyry	03, 07, 08, 09, 13	sediment layer	
Kaiwara ash flow tuff	01, 04, 06, 10, 11, 12, 18, 21, 22, 25, 26	Azuchi ash flow tuff	16, 20, 21

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Table 2. Site-Mean Directions of Characteristic Remanent Magnetizations and Virtual Geomagnetic Pole Positions From the Koto Rhyolite[a]

	Structural Attitude			In Situ		Tilt-Corrected				Virtual Geomagnetic Pole	
Site	Strike	Dip	N	Declination	Inclination	Declination	Inclination	k	α_{95}	Latitude	Longitude
					Ash Flow Tuffs						
KM01	183.5	15	15	61.2	41.1	51.4	53.1	215.3	2.6	36.2	221.1
KM02	18.5	25	9	88.8	57.2	95.9	33.1	23.0	11.0	21.5	194.4
KM06	...	...	7	79.5	65.3	...	...	84.3	6.8	31.6	187.7
KM10	91.5	22	10	98.5	53.9	123.8	46.2	247.8	3.1	13.0	193.1
KM11	143.5	40	7	72.0	50.5	153.0	78.2	138.9	5.1	31.0	207.8
KM12	143.0	12	8	65.6	51.3	70.2	62.9	123.9	5.0	36.2	209.6
KM19	191.5	50	9	104.9	56.9	275.2	73.0	20.7	5.8	10.5	187.6
KM21	263.5	30	8	70.6	76.1	17.3	54.5	22.9	11.8	39.5	169.1
KM25	...	...	5	91.7	67.2	...	...	78.1	8.7	25.1	181.6
KM26	299.5	14	4	81.1	43.8	72.7	34.3	78.7	10.4	21.3	209.2
KM27	...	...	9	84.5	48.2	...	...	19.2	12.1	20.4	204.3
KM28	...	...	9	129.3	53.2	...	...	47.4	7.6	−6.4	176.6
KM29	...	...	10	51.3	34.5	...	...	59.6	6.3	42.1	232.0
KM35	...	...	9	69.2	50.8	...	...	28.7	9.8	33.2	208.7
KM38	...	...	9	87.3	46.7	...	...	14.4	14.0	17.7	204.1
KM39	...	...	9	55.9	54.3	...	...	457.2	2.4	44.8	209.7
KH02	195.5	12	10	109.8	68.8	115.2	80.7	61.5	6.2	16.5	173.1
KH03	323.5	14	5	279.7	−53.4	269.4	−42.8	331.3	4.2	11.9	192.8
KH04	...	...	4	286.6	−36.4	...	...	84.5	10.1	−1.2	200.2
KH08	...	...	6	280.7	−44.5	...	...	106.2	6.5	6.7	198.7
KH11	...	0	5	258.3	−55.9	258.3	−55.9	87.5	8.2	28.4	199.6
KH13	...	0	5	254.7	−49.0	254.7	−49.0	40.3	12.2	28.3	207.8
KH16	83.5	13	7	147.8	53.6	153.4	41.6	91.3	6.3	−14.5	163.2
KH17	269.5	20	6	288.0	−34.0	273.6	−38.0	53.0	9.3	−3.2	200.6
					Intrusive Rocks						
KM05	...	...	8	229.9	−60.3	...	...	18.3	13.3	51.4	202.0
KM07	...	...	8	75.6	68.7	...	...	329.4	3.1	35.3	183.2
KM08	...	...	9	84.5	67.9	...	...	36.4	8.6	29.7	182.6
KM09	...	...	9	89.5	37.7	...	...	20.0	11.8	12.4	209.0
KM13	...	...	8	108.2	65.6	...	...	28.4	4.4	14.7	177.6
KM14	...	...	9	48.1	81.1	...	...	50.9	9.8	45.4	154.8
KM23	...	...	9	250.2	−65.2	...	...	50.9	7.3	37.6	190.0
KM24	...	...	9	32.5	66.8	...	...	20.7	11.6	62.3	185.2
KH07	...	...	6	74.6	64.0	...	...	84.3	7.3	34.3	190.7

[a] N, number of samples; k, Fisher's precision parameter; α_{95}, radius of cone of 95% confidence.

flow tuffs. Gentle tilting and the absence of regional trend suggest that postemplacement tilting is not so large for the ash flow tuffs and also for associated intrusive rocks in which we could not directly measure the structural attitudes. The Koto rhyolite is considered to have experienced no large regional tilting as a whole until the present time.

[14] Characteristic and secondary remanent magnetizations were isolated by stepwise thermal demagnetization by heating samples in a mu-metal-shielded oven (residual magnetic field <10 nT). Supplementary alternating field (AF) demagnetization was also performed using a custom-made AF demagnetizer with a triaxial tumbling device. Measurements of remanent magnetization were made on a Schonstedt spinner magnetometer (SSM-1A) and an ScT two-axis cryogenic magnetometer. To identify magnetic minerals, we conducted thermal demagnetization of orthogonal isothermal remanent magnetization (IRM) [*Lowrie*, 1990] for selected samples. When imparting IRM, a cylindrical specimen was set into a cubic holder to minimize orientation errors. For modified Lowrie-Fuller tests [*Johnson et al.*, 1975], anhysteretic remanent magnetization (ARM) was imparted using a modified AF demagnetizer with a small coaxial solenoid coil. All the above measurements were made at the paleomagnetics laboratory of Kyoto University.

[15] Magnetic components isolated by stepwise demagnetization were recognized from visual inspection of linear segments of vector end point diagrams. Characteristic remanent magnetization (ChRM) is defined by a linear segment trending toward the origin of the vector end point diagram. For samples showing multicomponent magnetizations, we performed stepwise demagnetization for all the samples collected at a single site. Directions of magnetic components were determined with principal component analysis for the observed linear segments [*Kirschvink*, 1980]. For single-component magnetizations ChRM directions were determined through a single-step demagnetization which was proved to give precise directions as principal component analysis [*Beck et al.*, 2001]. Site-mean directions were obtained by vector addition of each sample's characteristic magnetization, and the statistical parameters were given according to that in the work of *Fisher* [1953]. Pole positions were calculated by averaging virtual geomagnetic poles obtained from site-mean directions of individual sites.

4. Rock Magnetic Analysis

[16] To identify magnetic minerals of the Koto rhyolite, mutually perpendicular three-component IRMs were subject

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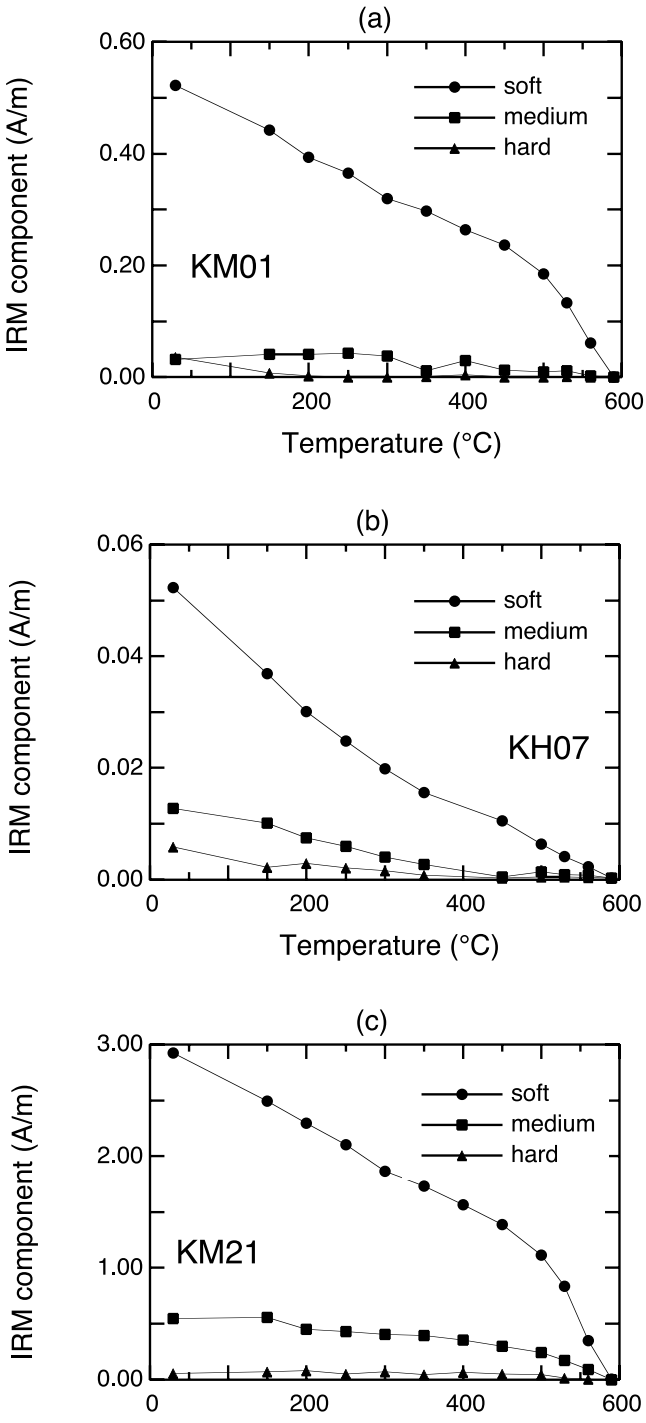


Figure 3. Thermal demagnetization of orthogonal isothermal remanent magnetizations (IRM) [*Lowrie*, 1990] for (a) ash flow tuffs, (b) intrusive rocks, and (c) contact aureole. Soft, medium, and hard denote the IRM components induced in the direct fields of 0.12, 0.4, and 1.3 T, respectively.

to stepwise thermal demagnetization up to 600°C. Typical results for ash flow tuffs and intrusive rocks are shown in Figure 3. For both ash flow tuffs and intrusive rocks, the softest components (induced in fields of 0.12 T) dominate over medium and hard components (induced at 0.4 and 1.3 T) and are completely demagnetized at 580°C. These results indicate that the dominant remanence carrier is nearly stoichiometric magnetite with the Curie temperature of 578°C. Medium and hard components were significantly small and also demagnetized at about 580°C, which is also indicative of magnetite and probably caused by minor orientation errors when magnetizing along three mutually perpendicular axes.

[17] The softest IRM component of ash flow tuffs is rapidly demagnetized within a temperature range of 100°C below 580°C (Figure 3a), whereas those of intrusive rocks are gradually demagnetized from room temperature to 580°C (Figure 3b). Even for IRMs, which are influenced more by multidomain grains than thermoremanent magnetizations (TRM) or ARMs, such high and narrow unblocking temperatures for the ash flow tuffs are suggestive of predominately single-domain or pseudo-single-domain magnetite. An ash flow tuff sample from the contact aureole shows an essentially identical result with that of nonbaked ash flow tuff (Figure 3c). This result shows that thermal metamorphism did not significantly change the magnetic mineralogy in the contact aureole, which implies that chemical remanent magnetization due to thermal metamorphism is not plausible.

[18] Single-domain type curves [*Dunlop*, 1983] were observed for ash flow tuffs and multidomain type curves for intrusive rocks in the modified Lowrie-Fuller tests [*Johnson et al.*, 1975]. IRM was imparted by a direct field of 1.0 T, and ARM was acquired at two levels of direct fields of 0.1 and 0.3 mT with a peak AF of 100 mT. AF demagnetization of ash flow tuffs shows initial slow decays especially for ARM, and ARM acquired at a direct field of 0.1 mT is most resistant to AF, while IRM is least resistant (Figure 4a). This type of behavior is typical of single domain, which is observed for single-domain and pseudo-single-domain grains up to 10–15 μm [*Bailey and Dunlop*, 1983] and suggestive of carrying stable natural remanent magnetization (NRM). For intrusive rocks, AF demagnetization curves of ARM and IRM have nearly exponential shapes, although showing slightly inflected curves (Figure 4b). IRM is most resistant to AF demagnetization, and ARM induced by a direct field of 0.3 mT is slightly harder than ARM of 0.1 mT. This is a multidomain-type behavior, which is indicative of multidomain grains >15 μm and is generally responsible for unstable NRM of many intrusive rocks [*Dunlop*, 1983]. It should be noted that the inflected curves suggest that minor hard fractions coexist with multidomain grains. An ash flow tuff sample from the contact aureole showed a single-domain-type behavior (Figure 4c) as found for nonbaked ash flow tuffs (Figure 4a). This result indicates that magnetic grain size in the contact aureole is not appreciably changed and hence support the view that chemical remanent magnetization is not plausible, which is expected from the unaltered magnetic mineralogy (Figure 3).

5. Demagnetization Results

[19] NRM intensities range from 10^{-2} to 10^{-4} A/m for ash flow tuffs, while intrusive rocks generally have lower NRM intensities from 10^{-4} to 10^{-5} A/m. Three or more specimens were selected as pilot samples and were subjected to stepwise thermal or AF demagnetization to analyze components of remanent magnetization. Generally, thermal demagnetization was found to be more effective for isolat-

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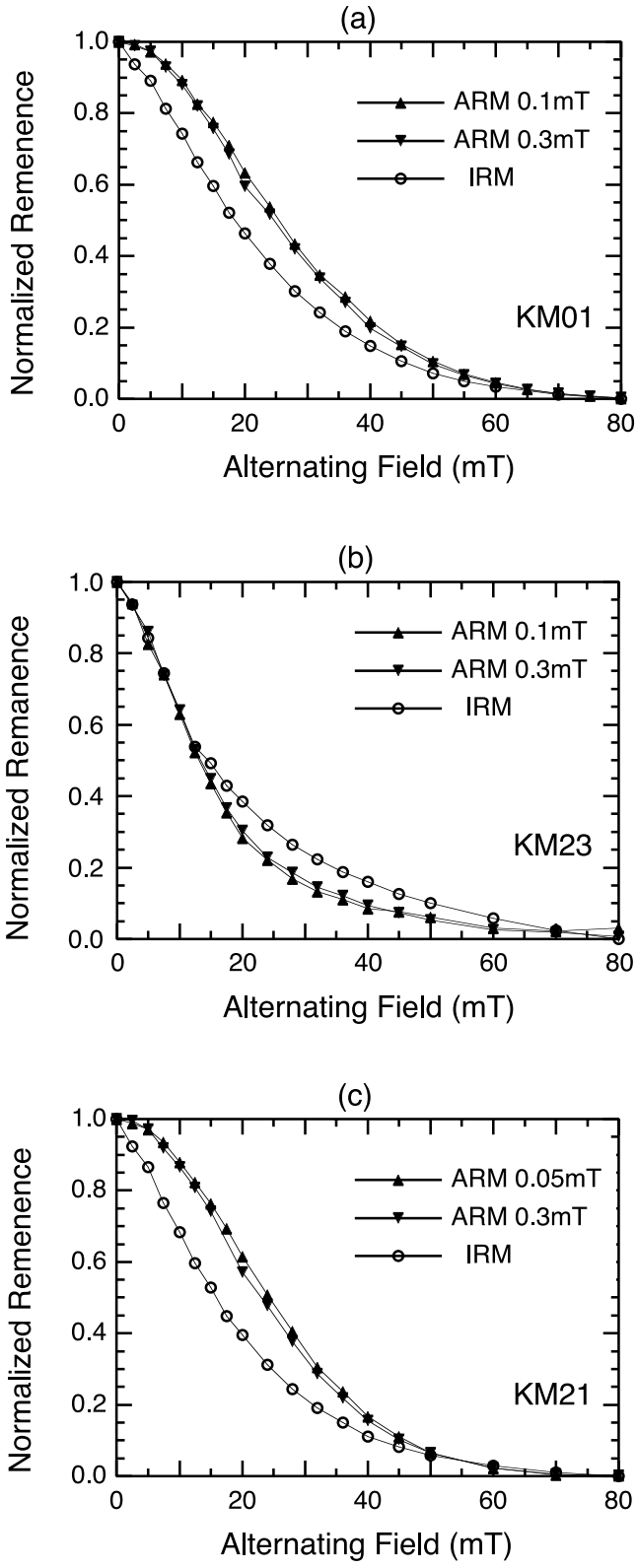


Figure 4. Alternating field demagnetization of anhysteretic remanent magnetization (solid triangles) and isothermal remanent magnetization (open circles) for (a) ash flow tuffs, (b) intrusive rocks, and (c) contact aureole.

ing ChRM (Figure 5). Multicomponent magnetizations, which were identified for several sites through thermal demagnetization, were never observed during AF demagnetization even when using sister specimens taken from a same hand sample. Therefore we employed thermal demagnetization to isolate characteristic magnetizations from secondary ones. Based on the magnetization behavior during thermal treatment, samples can be divided into three groups: (1) samples with a single-component magnetization of high (>500°C) unblocking temperatures; (2) samples in which its magnetization is unblocked over a wide range of temperature; and (3) samples with multiple components of magnetization. ChRM directions and related statistical parameters are summarized in Table 2.

[20] Most of ash flow tuffs exhibit essentially single component magnetizations with high unblocking temperatures (500°–580°C) (Figure 6a), which are typically observed in densely welded ash flow tuffs (e.g., Kaiwara ash flow tuff). Such a narrow unblocking temperature range of NRMs just below the Curie temperature of magnetite is consistent with the demagnetization results of IRM for ash flow tuffs (Figure 3a). Single-domain or pseudo-single-domain magnetite is responsible for the ChRM as indicated by rock magnetic analyses, which strongly suggest that ash flow tuffs faithfully retain stable ancient magnetizations. It is highly probable that ChRMs of ash flow tuffs would represent thermoremanences acquired during the initial cooling following emplacement or thermochemical remanence with the growth of magnetite during devitrification of the glass matrix. In addition to the ChRM, some ash flow tuffs, which primarily come from moderately welded ash flow tuffs (e.g., Yatsuoyama pyroclastics), carry low unblocking temperature (<300°C) components (Figure 5a). These secondary components may be viscous remanences acquired in the present geomagnetic field or during laboratory storage. To determine the site-mean ChRM directions, the rest of the specimens from the same site were subjected to thermal demagnetization at a temperature level at which ChRM was recognized on the vector end point diagram.

[21] For intrusive rocks, the unblocking temperature spans a wider and lower temperature range, and erratic behaviors are often seen in the high-temperature range (Figure 6b). ChRMs were not isolated by stepwise thermal demagnetization at many sites of intrusive rock. Viscous overprints parallel to the present geomagnetic field were often observed and in some cases dominate the NRM. Multidomain magnetite, as indicated by the rock magnetic analyses (Figures 3b and 4b), can account for these demagnetization characteristics of the intrusive rocks. Nevertheless, ChRMs were isolated at some sites. They apparently reside in single-domain or pseudo-single-domain grains coexisting with multidomain grains as shown by slightly inflected AF demagnetization curves of ARMs (Figure 4b).

[22] Multicomponent magnetization behavior was observed at several sites during thermal demagnetization. In addition to the ChRM, a secondary magnetization component with moderate unblocking temperatures (300°–500°C) was observed. In cases where pilot specimens exhibited multicomponent behavior, stepwise thermal demagnetization was applied to all the other specimens from the same site. Two examples of multicomponent behavior are discussed below.

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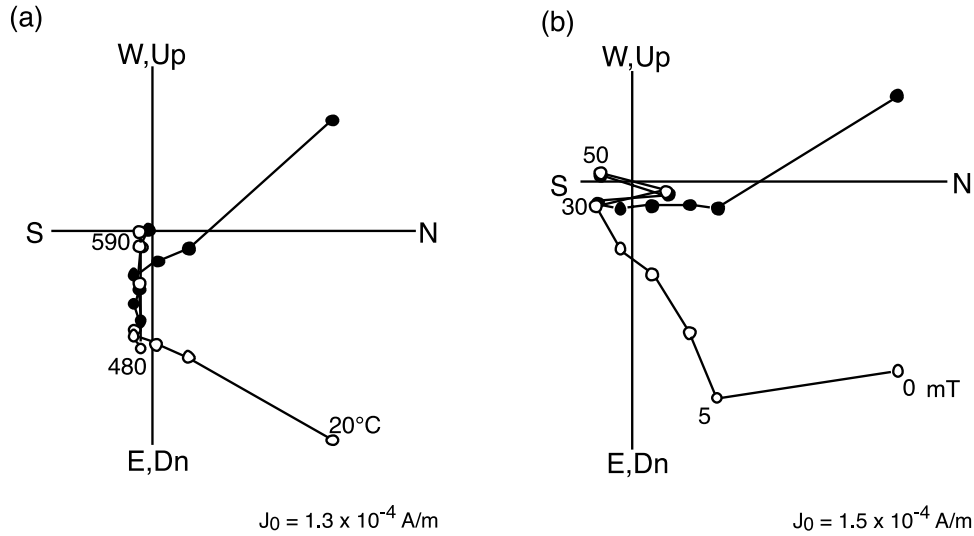


Figure 5. Vector end point diagrams to compare (a) thermal and (b) alternating field demagnetization results for sister specimens taken from a same hand sample of weakly welded ash flow tuff (site KM02).

[23] As an example, a thermal demagnetization result showing multiple magnetization components and their mean directions of each component are illustrated for site KH18 (Koshigoe ash flow tuff) (Figure 7), where the Kamewari ash flow tuff (site KH17) directly overlies the Koshigoe ash flow tuff (site KH18). Ash flow tuffs at site KH18 suffered thermal alteration due to the heat transferred from the overlying ash flow tuff. Three components of magnetization were found at site KH18 and were labeled A, B, and C with increasing unblocking temperature. The A component was unblocked between 320° and 420°C and was of normal polarity. The B component unblocked from 520° to 550° C, and the C component, which is equivalent to ChRM, was unblocked in a temperature range from 550° to 580°C. Both the B and C components show reversed polarities with nearly antipodal directions to the A component (Figure 7b). The overlying Kamewari ash flow tuff (KH17), being the heat source for the overprinting, shows a similar direction of reversed polarity with the B component. We suppose that the B component is a secondary magnetization overprinted by the overlying Kamewari ash flow tuff. The C component (ChRM) must predate the emplacement of Kamewari ash flow tuff and is highly probable to be the primary magnetization acquired at initial cooling. The A component of positive inclination is virtually antiparallel to the B or C components, and the direction is largely different from that

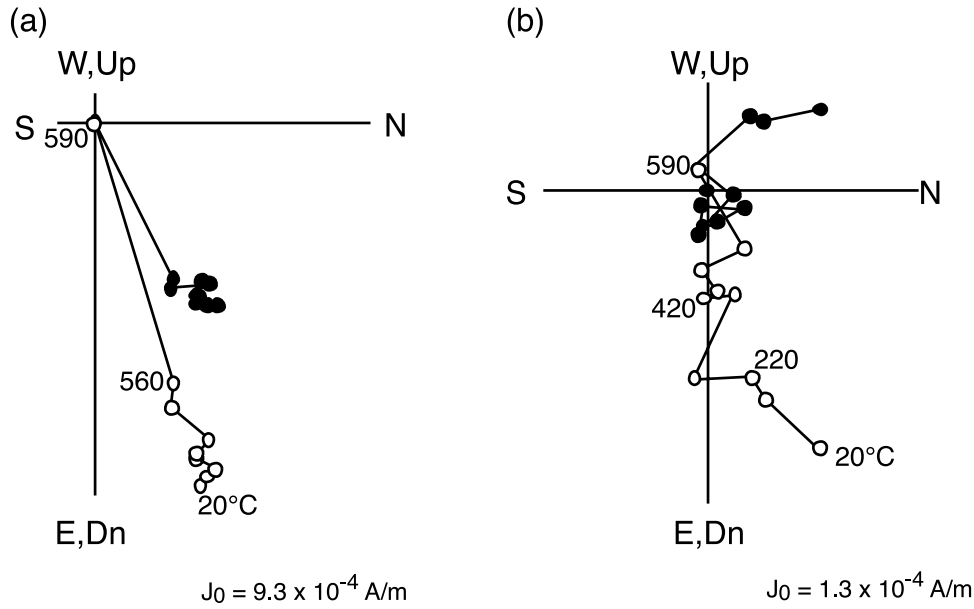


Figure 6. Vector end point diagrams representing thermal demagnetization results for (a) an ash flow tuff sample (site KM12) showing single-component magnetization of high (>500°C) and narrow unblocking temperature, and (b) an intrusive rock sample (site KM15) showing erratic behavior of magnetization with low and distributed unblocking temperature.

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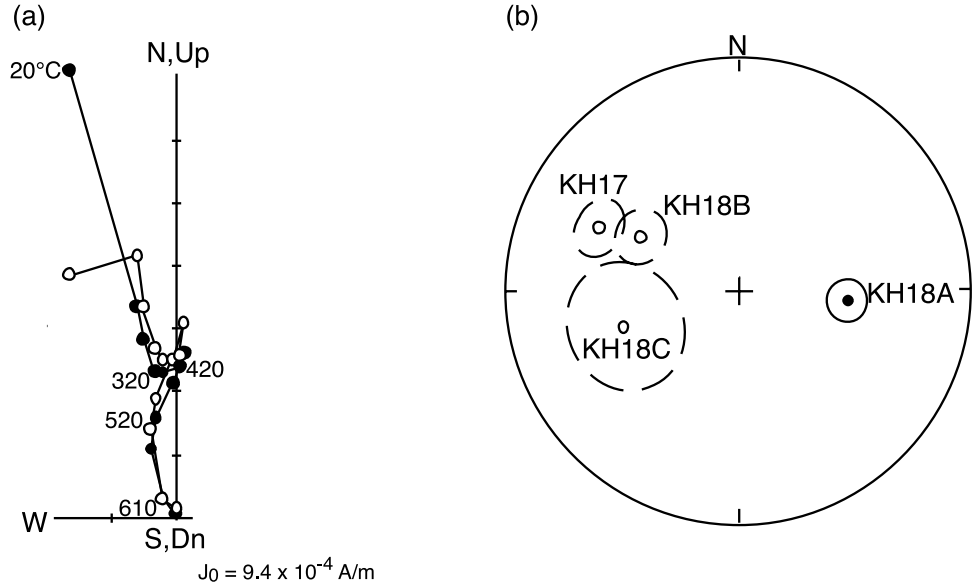


Figure 7. Multicomponent magnetizations revealed by thermal demagnetization. (a) Vector end point diagram showing a multicomponent behavior for site KH18. (b) Equal-area plot of multiple magnetization components of site KH18 (labeled as "KH18A," "KH18B," and "KH18C") and an overlying ash flow tuff (site KH17).

of the present geomagnetic field. It is obvious that the A component is also of ancient origin, but no other heat source which imprinted the A component was confirmed near the outcrop. Nonexposed or eroded-out intrusive rocks may have been a heat source for the secondary A component.

[24] Magnetizations of ash flow tuffs in the contact aureole also show multicomponent demagnetization behavior and systematically vary as a function of the distance from the contact (Figure 8). In the southern part of the main body, there is a ring dike of Inugami granite porphyry about 1 km in width, and it is associated with a baked contact zone in the country rock of Kaiwara ash flow tuff. The country rocks show positive inclinations at all sites except in the contact aureole, while the ChRM direction of the ring dike is almost antiparallel to those of the country rock. At the nearest site (KM22) to the contact, within a distance of less than 100 m from the contact (Figure 8a), the magnetization of the ash flow tuff is single-component and parallel to ChRM of the dike (Figure 8c). The magnetization was totally reset at the time of dike intrusion, and only the overprint component remains at the KM22. At two sites (KM21 and 25) located about 250 m away from the contact (Figure 8a), two mutually almost antiparallel components are observed during thermal demagnetization (Figure 8b). The high-temperature component (550°–580°C) is parallel to the ChRM of the nonbaked ash flow tuffs, and the medium-temperature (300° and 550°C) component is parallel to the ChRM of the dike (Figure 8c). The high-temperature component is interpreted to be the original magnetization acquired at initial cooling, whereas the medium-temperature component was acquired at the time of dike intrusion. Because there is no difference in magnetic mineralogy and grain size between the baked and nonbaked ash flow tuffs (Figures 3a, 3c, 4a, and 4c), it is highly probable that the medium-temperature magnetization in the contact aureole is of partial TRM origin. Low unblocking temperature component (<300°C) is parallel to the present geomagnetic field and is probably a viscous remanence.

[25] The contact test is a very powerful field test to constrain the timing of magnetization acquisition [*Everitt and Clegg*, 1962]. Along a profile crossing the contact aureole, we found a total reset zone adjacent to the contact and a partial reset zone showing multicomponent magnetizations further away from the contact (Figure 8). This positive contact test provides unambiguous evidence that ChRM of the country rock (ash flow tuffs) precedes the intrusion of the ring dike. Furthermore, we found multicomponent magnetizations in an ash flow tuff layer, which is thermally affected by a directly overlying ash flow tuff layer (Figure 7). This result imposes a severe constraint on the timing of ChRM acquisition of the lower ash flow tuff, which predates the emplacement of the upper ash flow tuff. Also, these contact tests suggest that the magnetizations of the ring dike or the upper ash flow tuffs are not affected by any subsequent remagnetization. Possibility of pervasive remagnetization for the Koto rhyolite could be eliminated.

6. Analysis of Paleomagnetic Directions

[26] Site-mean ChRMs of the Koto rhyolite were obtained through thermal demagnetization for 33 sites (Table 2). These ChRMs have the same polarity within each lithologic unit, and alternating polarities are observed according to the stratigraphic succession. In the main body, the Kaiwara ash flow tuff and the associated Hatasho quartz porphyry are normally magnetized except for the totally reset zone of the contact aureole. The Yatsuoyama pyroclastics, which overlie the Kaiwara ash flow tuff and a intervening sediment layer, also have a normal polarity magnetization. The inner ring dikes intruding the Kaiwara ash flow tuff exhibit reversed polarities, while the outer ring dikes are normally magnetized. Reversely magnetized ash

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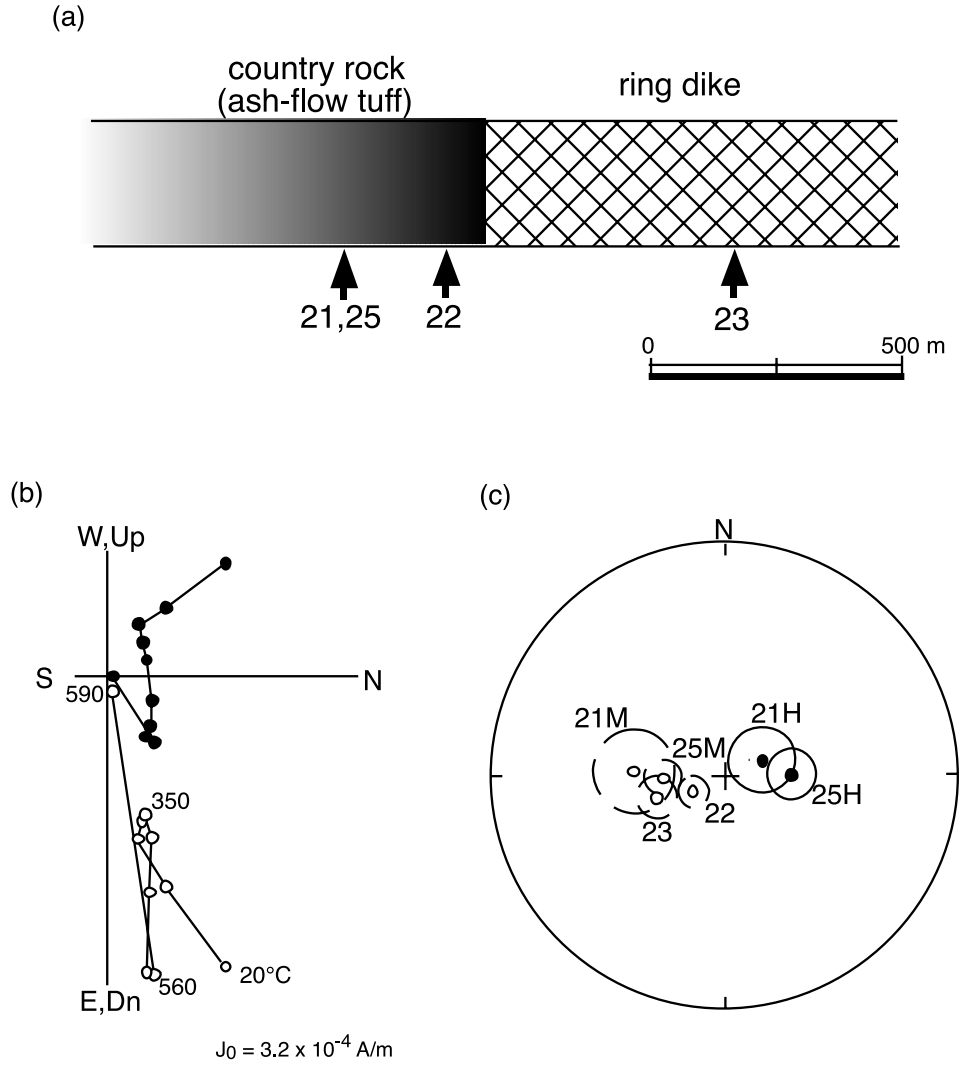


Figure 8. (a) Schematic profile of the contact aureole near the boundary of a ash flow tuff (site KM21, 22, and 25) and a ring dike (site KM23). (b) An example of multicomponent behavior during thermal demagnetization for a partly remagnetized baked ash flow tuff (KM21). (c) Equal-area plot of the magnetization components of baked ash flow tuffs and a ring dike. Labels such as "H" and "M" denote high (550°–580°C)- and medium (300°–550°C)-temperature components, respectively, for sites KM21 and KM25, and "22" and "23" show characteristic remanent magnetizations of sites KM22 and KM23, respectively.

flow tuffs are dominant in the smaller bodies, but Azuchi ash flow tuff that underlies the other ash flow tuffs, separated by a sediment layer, is magnetized normally. At one site of intrusive rocks in the smaller bodies, we found a normal-polarity magnetization. It is reasonable to assume that such a polarity sequence reflects geomagnetic polarity reversals, because the geomagnetic field was reversed fairly frequently in the Late Cretaceous [*Ogg*, 1995]. Since emplacement of ash flow tuffs or dikes occurs interruptedly, it is difficult to assign any lithologic unit to a particular polarity epoch and to correlate the lithologic units between the main body and the smaller bodies. Nevertheless, stratigraphically constrained polarity sequence can exclude the possibility that the entire region of the Koto rhyolite suffered from pervasive remagnetization after emplacement or intrusion. It is strongly suggested that the Koto rhyolite preserves the original magnetization acquired at or shortly after emplacement or intrusion.

[27] ChRMs of normal polarity were found at 25 sites and of reversed polarity at 8 sites (Figure 9), and the site-mean ChRMs of normal polarity as well as of reversed polarity are well clustered in the in situ reference frame though the reversed directions show a somewhat elongated (i.e., non-Fisherian) distribution. The mean normal direction (D = 82.6°, I = 59.3°, k = 21.3, and α_{95} = 6.4° for 25 sites) and the mean reversed direction (D = −90.3°, I = −51.4°, k = 23.2, and α_{95} = 11.7° for 8 sites) are almost antipodal to each other and pass a reversal test with a B classification [*McFadden and McElhinny*, 1990]. This positive reversal test indicates that the contamination of ChRM by other magnetization components is negligibly small. Also, these site-mean directions are largely deflected to the east from

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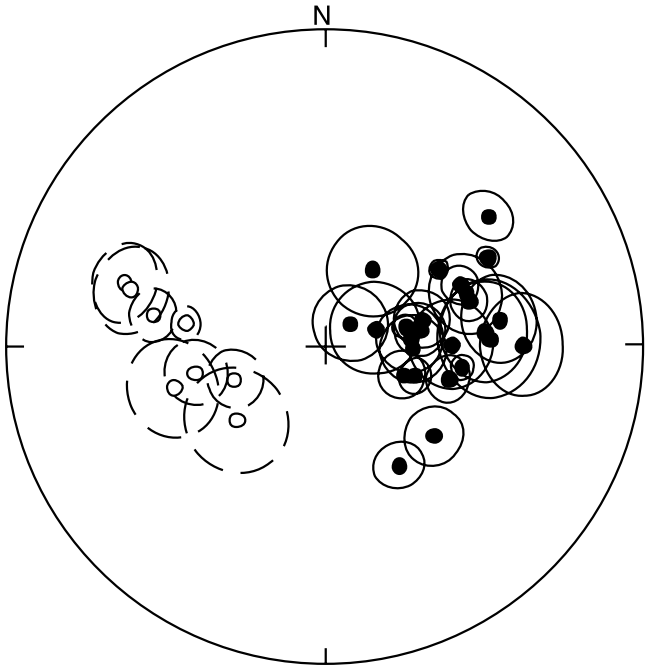


Figure 9. Equal-area projection of in situ site-mean directions with 95% confidence circles of characteristic remanent magnetizations. Solid/open symbols represent directions plotted onto lower/upper hemispheres.

either the present geomagnetic field or the geocentric axial dipole field so that the ChRM should represent a reliable ancient magnetization direction.

[28] Directions of ChRM at 14 sites of ash flow tuffs were corrected based on the eutaxitic structures observed at outcrops (Figure 10). This tilt correction results in an increase of dispersion as shown by a significant decrease of k (k_2/k_1 = 0.379, where k_1 and k_2 are the precision parameters [*Fisher*, 1953] before and after tilt correction, respectively). This negative tilt test indicates that the ChRMs of ash flow tuffs were acquired after tilting and should be considered in the in situ reference frame. However, the clearly defined polarity sequence, the positive reversal test, and the positive contact test rule out the possibility of pervasive remagnetization. Acquisition of ChRM would shortly follow collapse and tilting of ash flow tuffs within the Koto rhyolite cauldron, which is recorded as bedding attitudes of eutaxitic structures. The orientations of eutaxitic structure of ash flow tuffs are almost random, and their average is nearly horizontal, suggesting that the overall area of the Koto Rhyolite has not suffered any regional tilting until the present time. Therefore an overall mean of 33 site-mean ChRM (D = 84.6°, I = 57.5°, α_{95} = 5.5°) of ash flow tuffs and intrusive rocks should yield the time-averaged dipole field of ancient age which averages out the secular variation spanning several geomagnetic epochs. By averaging virtual geomagnetic poles calculated from each site-mean ChRM, we obtained a paleomagnetic pole (194.1°E, 25.6°N, α_{95} = 7.1°) from the Koto rhyolite.

7. Discussion

7.1. Origin and Age of Magnetization

[29] There are three possibilities for the origin of the ChRM of the ash flow tuffs of the Koto rhyolite: a primary thermoremanent magnetization, a thermochemical remanent magnetization acquired at initial cooling following emplacement, or a thermochemical remanent magnetization during epithermal or hydrothermal alteration in the Koto rhyolite cauldron. The ChRM of the ash flow tuffs is predominantly carried by nearly pure magnetite of fine (single- or pseudo-single-domain) grain size, which was shown by rock magnetic analyses and thermal demagnetization of NRM (Figures 3a, 4a, and 6a). Fine-grained magnetite grown through devitrification of volcanic glass in ash flow tuffs [*Schlinger et al.*, 1988] should be the carrier of the ChRM.

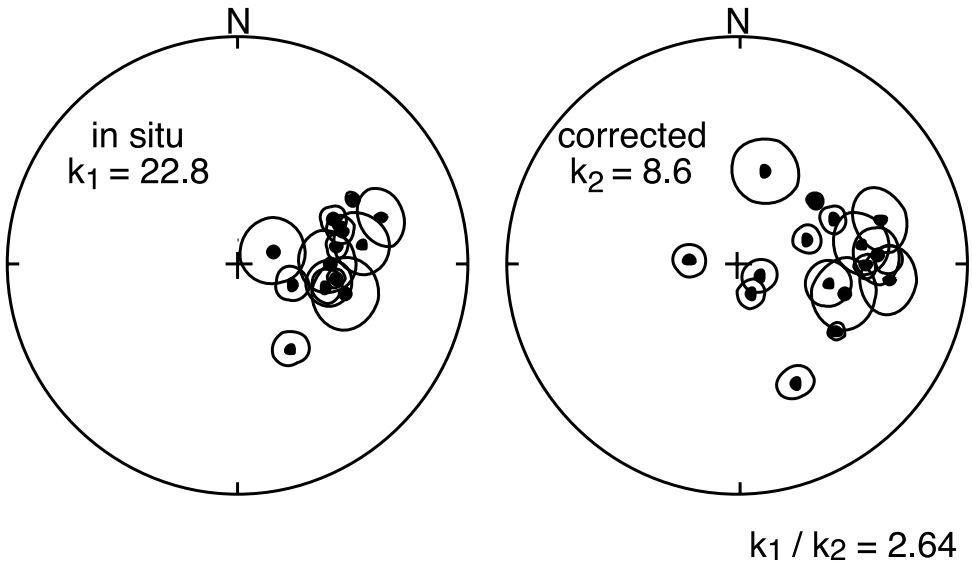


Figure 10. Equal-area projection of in situ and tilt-corrected site-mean directions with 95% confidence circles for ash flow tuffs. Reversed magnetizations are converted to the antipodal counterpart to demonstrate the clustering of the data.

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Ash flow tuffs are commonly emplaced at temperatures between 500° and 700°C [*Smith*, 1960]. If the magnetite was grown above the Curie temperature of magnetite (578°C), a primary TRM would be acquired. However, the negative tilt test rules out a possibility of the pretilting magnetization and hence a primary TRM. On the other hand, the ash flow tuffs and the intrusive rocks of the Koto rhyolite have suffered alteration to some extent. Although some localized alteration might have partly affected the magnetization, it is unlikely that extensive remagnetization covering the whole area of the Koto rhyolite occurred in a short period, because normal and reversed ChRM were observed to be constrained stratigraphically. Therefore a thermochemical remanent magnetization acquired soon after initial cooling is the most plausible explanation for the posttilting ChRM of the Koto rhyolite. A posttilting thermochemical remanence has been previously reported from ash flow tuffs formed in a cauldron accompanying ring dikes [*van Fossen and Kent*, 1990]. It is interpreted that the ChRM of the Koto rhyolite is carried by authigenic magnetite grown below the Curie temperature and was acquired shortly after collapse and tilting of ash flow tuffs within the Koto rhyolite cauldron.

[30] We need to assign an age to the magnetization of the Koto rhyolite for further discussion. A series of paleomagnetic field tests confirmed that acquisition of the ChRM of the ash flow tuffs does not precede the tilting but predates the intrusion of the associated ring dikes. These results indicate that the ChRM is of ancient origin and provides a severe constraint on the magnetization acquisition in a narrow time range between the tilting of the ash flow tuffs and the dike intrusion. There is no significant difference in the published K-Ar ages of the ash flow tuffs and the ring dikes [*Sawada and Itaya*, 1993], so we suppose that the overall average of the various rock types represents the age of emplacement of the Koto rhyolite. The biotite K-Ar ages (closure temperature ~300°C) of the Koto rhyolite are concentrated at about 70 Ma [*Sawada and Itaya*, 1993], and the zircon fission track ages (closure temperature ~240°C) ranging from 60 to 70 Ma [*Ito*, 1989] suggests that no thermal event exceeding 240°C took place later than 60 Ma. The ChRMs have high unblocking temperature typically above 500°C. Any low-temperature (<240°C) thermal event would not significantly alter the remanence direction, although unblocking temperature is somewhat lowered when thermal effects are prolonged in natural conditions [*Pullaiah et al.*, 1975]. Therefore the range of 60–70 Ma of zircon fission track ages gives a lower limit to the age of the ChRM, and presumably such high-temperature magnetizations predate the closure of biotite K-Ar ages. We assume a nominal age of 70 Ma in Late Cretaceous for the paleomagnetic pole of the Koto rhyolite.

[31] Widespread remagnetization events have been suspected to occur in east Asia including China, Korea, and Japan [e.g., *Wang and Van der Voo*, 1993]. *Kent et al.* [1987] reported that Cretaceous to recent overprinting is pervasive in eastern China. *Lee et al.* [1987] noted that Early Cretaceous sediments from the Gyeongsang basin in South Korea are extensively overprinted in recent times. Moreover, *Moreau et al.* [1987] reported a negative tilt test for Cretaceous sediments (the Miyako Group) from northeast Japan and cast doubt on the previous paleomagnetic poles of pre-40 Ma rocks obtained from the entire region of Japan. Although no casual relationship is confirmed for these remagnetization events in east Asia, the importance of field tests was stressed to obtain reliable paleomagnetic data from such a tectonically active region.

[32] Some paleomagnetic studies argue based on field tests against widespread remagnetization. At a sampling site adjacent to the Miyako Group, *Otofuji et al.* [1997] found secondary magnetizations whose in situ directions are similar to those of the remagnetized component of the Miyako Group [*Moreau et al.*, 1987] in addition to high-temperature primary magnetizations, which pass the tilt test. In southwest Japan, Paleozoic limestones were reported to be pervasively remagnetized, which is supported by the uniform single-polarity (normal) magnetizations [*Morinaga et al.*, 1989]. However, the surrounding welded tuffs yielded stable paleomagnetic directions of dual polarity [*Otofuji and Matsuda*, 1987]. These examples suggest that remagnetization events have occurred in some areas of the Japan Arc, but that such events are quite limited in regional extent and not so intensive as to overprint surrounding rocks completely. A hypothesis that an intense widespread remagnetization was strong enough to erase original magnetizations [*Moreau et al.*, 1987] is not currently supported. Our detailed paleomagnetic analyses of the Koto rhyolite also eliminate the possibility of widespread remagnetization since we used a combination of several paleomagnetic field tests, including a baked contact test.

7.2. Tectonic Implications

[33] An apparent polar wander (APW) path for the last 100 Ma obtained from the San'in area of western part of southwest Japan [*Otofuji and Matsuda*, 1987] is used for comparison with the paleomagnetic pole from the Koto rhyolite. APW paths are usually not acquired in active continental margins, where tectonostratigraphic terranes have experienced differential displacement or rotation. However, the paleomagnetic data of the San'in area come from well-dated autochthonous igneous rocks covering a long time period from the Cretaceous to the Miocene in a relatively limited area (150 km × 50 km). We anticipate that the APW path should represent an APW path for the western part of southwest Japan. The 70 Ma Koto paleomagnetic pole is compared with the San'in areas 60 and 80 Ma paleopoles, which are averaged for the time intervals of 58–63 Ma and 80–92 Ma, respectively. Three 95% confidence ovals associated with these paleopoles are overlapped (Figure 11a), suggesting that these three paleopoles are statistically indistinguishable at 95% confidence level. This comparison of paleomagnetic poles within southwest Japan indicates that there was no significant relative tectonic motion between the western (San'in area) and eastern (Koto area) part of southwest Japan since the Late Cretaceous.

[34] This result provides further support for the viewpoint that the Koto and San'in areas were included in a single microplate during the Miocene clockwise rotation of southwest Japan. Virtually identical paleomagnetic directions from coeval rock units, covering the pre- and postrotation periods, were obtained from San'in area in the west to the Ichishi area in the east [*Hayashida*, 1994; *Otofuji et al.*, 1985]. This region extends about 500 km in the middle part of southwest Japan, which can be considered to have behaved as a single microplate during the Miocene rotation.

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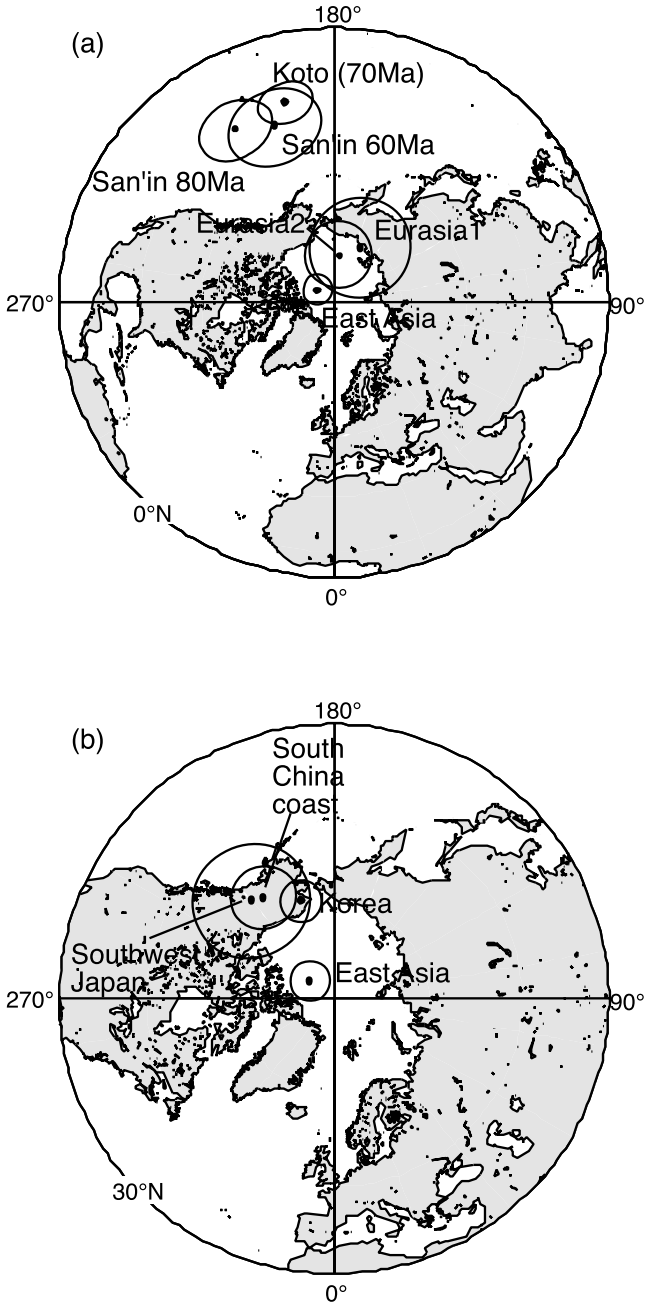


Figure 11. (a) Late Cretaceous paleomagnetic poles from San'in (*Otofuji and Matsuda* [1987], "San'in 60Ma" and "San'in 80Ma"), and Koto ("Koto (70Ma)") areas in southwest Japan, compared with coeval reference poles of Eurasia ("Eurasia1": *Irving and Irving* [1982], "Eurasia2": *Besse and Courtillot* [1991], and the east Asia's pole ("East Asia") calculated from the paleomagnetic data [*Otofuji et al.*, 1995; *Zhao et al.*, 1994]. (b) Paleomagnetic poles from the continental interior of east Asia, and Korea [*Lee et al.*, 1987; *Otofuji et al.*, 1986, 1983] and south China coast [*Gilder et al.*, 1993], and restored southwest Japan.

Pre-Neogene geologic terranes are juxtaposed parallel to each other, and the Median Tectonic Line (MTL) extends as a virtually straight line in the middle part of southwest Japan (Figure 1), which supports the viewpoint that no differential rotation occurred in the middle part of southwest Japan. In the eastern and western extremes of southwest Japan, however, smaller amounts of rotations were detected from slightly deflected paleomagnetic declinations [*Ishikawa*, 1997; *Itoh*, 1988; *Otofuji et al.*, 1999]. Bending of the MTL at the eastern and western extremes of southwest Japan (Figure 1) should be responsible for these paleomagnetic declinations observed at eastern and western extremes of the southwest Japan. Especially in the eastern extremes, the directions of nested parallel shear zones [*Kanaori*, 1990] are largely deflected compared to those of the middle part of southwest Japan.

[35] Paleomagnetic data of the Koto rhyolite gives a crucial constraint on the rotational mode of southwest Japan during the Miocene. *Jolivet et al.* [1995] postulated that the panel rotation along shear zones at least partly contributes to the Miocene rotation of southwest Japan in addition to the microplate rotation. If this is the case, the discordant directions of shear zones should be reflected in the observed paleomagnetic directions. The Koto area is located in a transitional region of shear zone directions (Figure 1) from NE-trending in the San'in area to NNW-trending in the eastern extreme. However, no significant relative rotation is detected between Koto and San'in areas from the paleomagnetic data (Figure 11a). This means that the panel-rotation along the shear zones is not related to the Miocene tectonic rotation of southwest Japan. Microplate rotation of the middle part [*Otofuji*, 1996] and ductile deformation of the eastern and western extremes [*Itoh and Ito*, 1989] are plausible in association with the Miocene rotation. Nevertheless, this does not necessarily exclude the possibility that the panel rotation occurred prior to the Miocene rotation. *Kanaori* [1990] presumed that the panel rotation has occurred mainly in the Late Cretaceous and the Paleogene.

[36] Reliable reference poles from continental interiors are a prerequisite for paleomagnetic studies of active continental margins, such as the Pacific coast of east Asia. Refinements of paleomagnetic poles from east Asia during the last decade have enabled us to discuss the continental margin deformation with more certainty. In previous syntheses of apparent polar wander paths for major continents [e.g., *Besse and Courtillot*, 1991; *Irving*, 1977; *Irving and Irving*, 1982], Eurasia, extending from Europe to east Asia, was assumed to be a single rigid continent, and the paleomagnetic poles were calculated by averaging the poles from several locations in Europe and Asia. However, there is a significant statistical discrepancy between the poles from east Asia and Europe for both the Cretaceous and Paleogene periods [*Otofuji et al.*, 1995]; the east Asia poles are near the present geographic pole, whereas the Europe poles are located at a lower latitude. The discordance of poles from Europe and east Asia suggests that the Eurasian continent did not behave as a rigid continent later than the Cretaceous period. In addition, the synthetic Eurasia poles are statistically weighted for Europe because most of the paleomagnetic data of Eurasia come from Europe. Therefore the Eurasia poles of *Irving* [1977] and *Besse and Courtillot* [1991], although these poles have been frequently cited in paleomagnetic studies of east Asia, are not appropriate as reference poles for discussing the continental margin deformation. Alternatively, we would regard the average of the Cretaceous paleomagnetic poles (233.5°E,

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83.3°N, α_{95} = 4.3°), obtained from north China, Mongolia, and Sikhote Alin [*Otofuji et al.*, 1995; *Zhao et al.*, 1994], as a reference pole for the continental interior of east Asia.

[37] We can detect a pre-Miocene rotation of southwest Japan in addition to the Miocene rotation by using the refined east Asian paleomagnetic pole as a reference pole. The pre-Miocene rotation of southwest Japan was considered to be negligible based on the APW path from San'in area [*Otofuji and Matsuda*, 1987]. However, this conclusion is not valid because it was based on a comparison with the apparent polar wander path of *Irving* [1977]. Now we calculate a combined pole of Late Cretaceous age for southwest Japan by averaging the Koto (70 Ma) and San'in (60 and 80 Ma) poles. To remove the effect of the Miocene clockwise rotation, then, we restored the combined pole position by rotating it counterclockwise about the Euler pole of 129°E, 34°N by 42°, which is calculated from the Miocene rotation by *Otofuji* [1996]. We plotted this restored southwest Japan pole (219.7°E, 63.0°N, α_{95} = 12.2°) of Late Cretaceous age, together with coeval poles from Korea (199.5°E, 68. 1°N, α_{95} = 4.4°) [*Lee et al.*, 1987; *Otofuji et al.*, 1986, 1983], and the south China coast (215.0°E, 63.7°N, α_{95} = 6.7°) [*Gilder et al.*, 1993] of the continental margin of east Asia (Figure 11b). These poles are concordant with each other but significantly different from a Cretaceous reference pole for the interior of east Asia [*Otofuji et al.*, 1995]. The amount of rotation relative to the continental interior of east Asia is calculated to be 24.6° ± 12.3° clockwise for southwest Japan, 20.0° ± 6.3° clockwise for Korea, and 20.8° ± 6.7° clockwise for the south China coast. No significant poleward displacements were detected for these three regions, suggesting that these rotations occurred along vertical or near-vertical axes. These observations imply that a significant pre-Miocene clockwise rotation of southwest Japan occurred in addition to the well-documented Miocene rotation and that such tectonic rotations are pervasive for the eastern continental margin of Asia. Some previous paleomagnetic studies on Mesozoic rocks suggested clockwise tectonic rotations in Korea [*Doh and Pier*, 1994; *Zhao et al.*, 1999] and the north China block [*Ma et al.*, 1993; *Uchimura et al.*, 1996], although the ages of rotation are not well constrained or somewhat inconsistent with our study.

[38] Large strike-slip faults extend along east Asia continental margin with an NE-SW trend as part of a strike-slip fault belt of about 1500-km width, including the Shi-Hang Fault in south China coast, the Kongju-Kwanju fault in Korea, and MTL in Japan [e.g., *Xu et al.*, 1987]. These faults were concurrently active in the Late Cretaceous with the formation of some sedimentary basins along the faults and were reactivated in the Paleogene [*Otsuki*, 1992]. These large faults might induce a local tectonic rotation in the vicinity [*Gilder et al.*, 1993], but we do not consider that such a local rotation could cause a coherent clockwise rotation as observed for the extensive continental margin from south China coast to southwest Japan. Large strike-slip faults usually bound minor faults forming parallel structural panels, which rotate to accommodate the shear by major strike-slip faults [e.g., *Nur et al.*, 1986]. *Kanaori* [1990] postulated that such a tectonic rotation due to structural panels occurred in southwest Japan in the Cretaceous and the Paleogene. Paleomagnetic results from southwest Japan, Korea, and south China coast indicate that such panel rotations took place in an extensive region of the eastern continental margin of Asia later than the Late Cretaceous.

[39] Cenozoic tectonic deformation in east Asia was regarded as a consequence of the collision of India against Eurasia and the extrusion of tectonic blocks [*Molnar and Tapponnier*, 1975]. In western China, tectonic blocks such as Tibet, Tarim, and Junggar are shortened or rotated as predicted by extrusion tectonics [e.g., *Cogne et al.*, 1996]. Also, in Indochina, numerous paleomagnetic data indicate that a large-scale clockwise rotation (~35°) occurred with respect to stable Eurasia since the Cretaceous [e.g., *Sato et al.*, 1999]. The Cenozoic collision of India certainly governs Cenozoic tectonic movements in the region near India. In the more eastern region, paleomagnetic poles from the north and south China blocks are concordant with each other in the Cenozoic, suggesting that the north and south China blocks have remained undisturbed, and no large-scale eastward displacement of these blocks was induced by India-Eurasia collision [*Zhao et al.*, 1994]. Sikhote Alin has also not been tectonically disturbed since the Paleogene [*Otofuji et al.*, 1995]. Further east, it might be expected that the tectonics of the eastern continental margin of Asia are influenced by the propagation of extrusion tectonics beyond the undisturbed north and south China blocks. However, oblique subduction of the Kula or Pacific plate with a westward component and the associated shear motion must be also considered in the convergent margin. The observed coherent rotations extending from southwest Japan to south China coast suggest that the subduction of the Kula or Pacific plate provides a substantial resistance to the eastward extrusion, and these two effects are reconciled along the eastern continental margin of Asia.

[40] At continental margins either tectonic processes of asymmetrical extension or dextral shear can lead to tectonic rotations defined by paleomagnetic data. Tectonic rotations covering the extensive Pacific coast of North America were explained with a consistent model that asymmetric extension and dextral shear are subequal contributors to the total rotations [*Wells and Heller*, 1988]. In the eastern margin of Asia, relative contributions of asymmetrical extension and dextral shear are variable along the margin. The 70° total rotation of southwest Japan since the Late Cretaceous is principally (~50°) attributed to the Miocene back arc extension of the Japan Sea [*Jolivet et al.*, 1995; *Otofuji*, 1996]. The remaining 20° rotation is concordant with those in Korea and south China coast (Figure 11b), where no large-scale extension is recognizable in the geological structures. Large strike-slip faults and accompanying structural panels are responsible for the consistent rotation along the east Asia margin (Figure 1); in addition, asymmetric extension gives rise to a larger amount of rotation in a localized area of the continental margin (i.e., southwest Japan). Extrusion due to India-Eurasia collision and oblique subduction of the Kula or Pacific oceanic plate could operate behind both the tectonic processes in the eastern margin of Asia.

8. Conclusions

[41] A Late Cretaceous (70 Ma) paleomagnetic pole was obtained from the Koto rhyolite in the eastern part of southwest Japan. Several paleomagnetic field tests con-

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firmed that the remanent magnetizations are not the product of any subsequent widespread remagnetization. The pole position is statistically indistinguishable from the coeval pole positions of the western part of southwest Japan. This fact implies that there has been no differential rotation in the middle part of southwest Japan since the Late Cretaceous and gives a further support to a microplate model for the Miocene rotation. The restored Late Cretaceous pole of southwest Japan, in which the effect of the Miocene rotation is eliminated, is still statistically different from the pole of the continental interior of east Asia. Cretaceous rocks from Korea and south China coast also show similar paleomagnetic poles with southwest Japan. About 20° clockwise rotation occurred for the extensive region of the eastern continental margin of Asia since the Late Cretaceous. Probably a shear motion along the continental margin is responsible for the observed rotation, which results from the interaction of the extrusion due to India-Eurasia collision and the subduction of the Kula or Pacific plate.

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